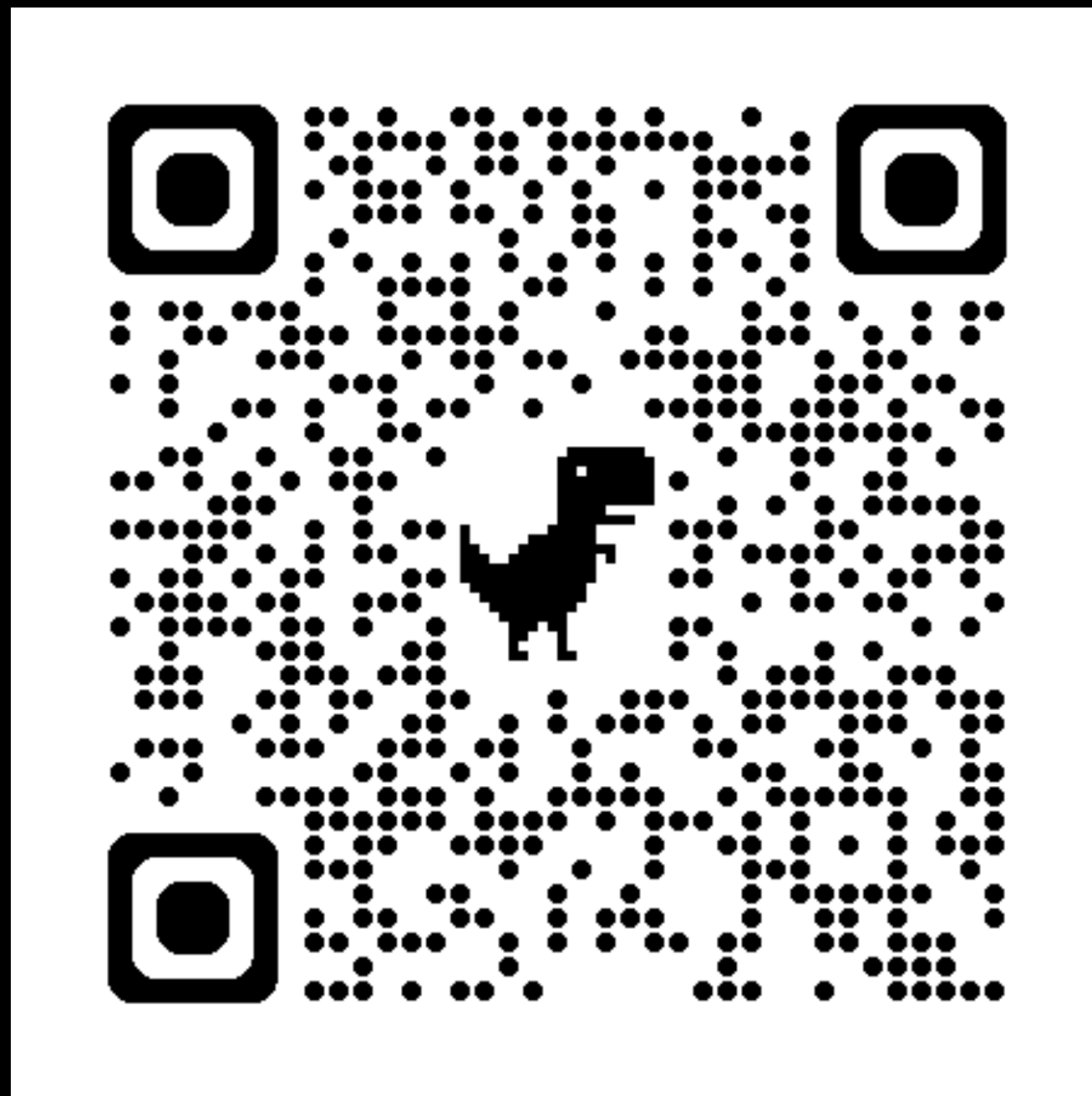


Takeaways

- Sinkhorn divergences - a debiased notion version of entropic OT.
 - Interpolate between OT and MMD: computationally, statistically, geometrically!
 - Essentially same computational complexity as EOT, but better behaved
- Barycenters = “averages of distributions” in Wasserstein geometry.
 - Classical: exact but expensive.
 - Entropic: scalable and widely used in ML.
 - Applications span vision, NLP, biology, and federated learning.

Interactive Demo: MNIST Barycenters



1. Import relevant function from POT to compute exact barycenter
2. Import relevant function from POT to compute sinkhorn barycenter
3. Play around with different digits, ϵ values

tinyurl.com/cs2840-lecture7